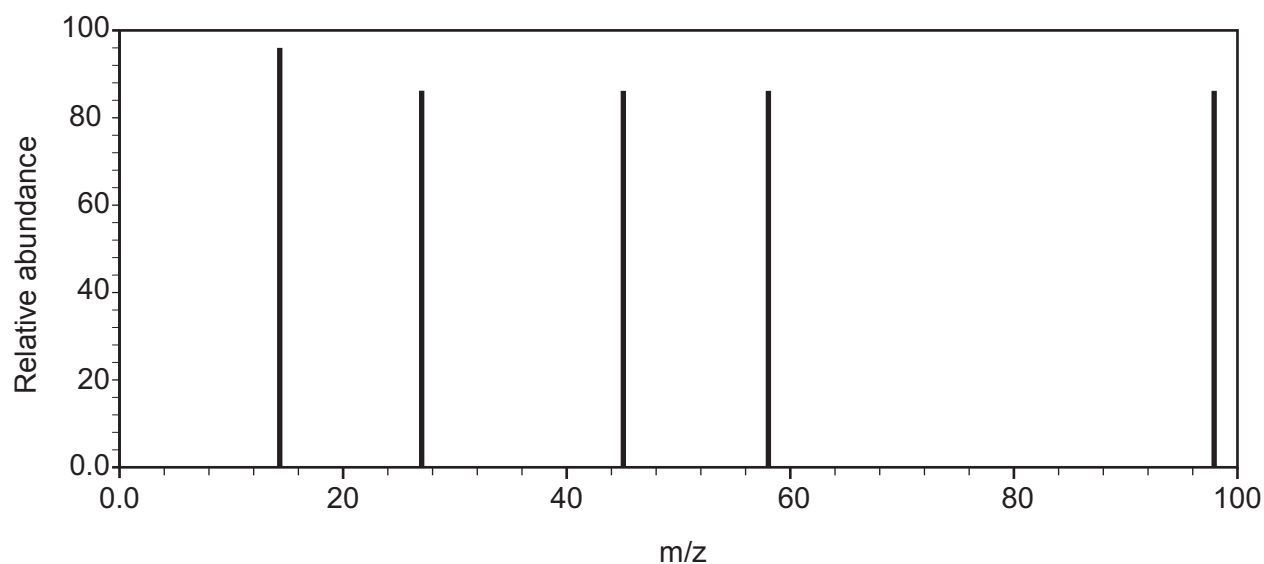


12. Chemists are investigating an unknown compound **X**. They obtain information from a variety of sources.

Compound **X** contains 61.2 % carbon, 6.1 % hydrogen and 32.7 % oxygen by mass. A simplified mass spectrum of compound **X** is shown below.



When solid sodium carbonate is added to an aqueous solution of **X** effervescence is observed.

There are 5 peaks in the ^{13}C NMR spectrum of **X**.

1 mol of **X** reacts completely with 320 g of bromine in the dark.



Use all the data given to find the structure of compound **X**. Explain what information can be found from each piece of data. [10]

Examiner
only

.....

.....

.....

.....

.....

10

END OF PAPER

